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United States Patent Application Publication

20250286565

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

BALL MARKS; Simeon Michael et al.

QUASI-CYCLIC LDPC CODES BASED ON GENERALISED QUADRANGLES

Abstract

In an aspect, digital communication coding methods (or simply coding methods) are provided, said methods comprising providing families of quasi-cyclic low-density parity check, LDPC, codes each represented by a single equation. Said single equation is derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes and allows quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and exceeding four hundred thousand bits. The construction of such check matrices is not possible by any other known method. The LDPC codes have different lengths and different rates and the families of LDPC codes are provided with a guarantee of minimum distance. The geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates. Systems, computing systems and computer programs suitable to perform such coding methods are also provided.

Inventors:	BALL MARKS; Simeon Michael (Castelldefels, ES), ORTEGA SÁNCHEZ COLOMER; Tomás (Sant Cugat Del Vallès, ES)
Applicant:	UNIVERSITAT POLITÈCNICA DE CATALUNYA (Barcelona, ES)
Family ID:	81653499
Appl. No.:	18/863823
Filed (or PCT Filed):	May 12, 2023
PCT No.:	PCT/EP2023/062797

Foreign Application Priority Data

EP	22382467.3	May. 13, 2022
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Publication Classification

Int. Cl.: H03M13/11 (20060101)

U.S. Cl.:

CPC H03M13/116 (20130101); H03M13/1105 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY [0001] This patent application claims priority from International Patent Application No. PCT/EP2023/062797 filed May 12, 2023, which claims priority to European Patent Application EP22382467.3 filed May 13, 2022. Each of these patent applications are herein incorporated by reference in its/their entirety. [0002] The present disclosure relates to digital communication coding methods and to systems, computing systems and computing programs suitable for performing such methods. [0003] The present disclosure further relates to methods, systems, computing systems and computing programs to encode and decode data in digital communications.

BACKGROUND

[0004] As known to the person skilled in the art a low-density parity-check (LDPC) code C is a linear code determined by a sparse parity-check matrix H having a small number of 1s per column. The parity-check matrix H may be represented by a bipartite Tanner graph wherein each column of H is represented by a transmitted variable node, each row by a check node, and each “1” in H by a graph edge connecting the variable node and check node that correspond to the column-row location of the “1”. Each check or constraint node defines a parity check operation. The fraction of a transmission that carries information is called the rate of the code. An LDPC code may be encoded by deriving an appropriate generator matrix G from its parity-check matrix H . An LDPC code may be decoded efficiently using well-known iterative decoding algorithms that pass messages along edges of the code's Tanner graph from variable nodes to check nodes and vice-versa until convergence is obtained. However, for very long codes, such iterative processes may be costly and lower complexity algorithms are also well known, which do perform efficiently for certain long LDPC codes in which the column weight is relatively high.

[0005] A quasi-cyclic low-density parity-check matrix is completely determined by a reduced matrix $H_{\text{sub.rep}}$ whose (i, j) entry $(H_{\text{sub.rep}})_{\text{sub.ij}}$ consists of either the empty set or a small subset of numbers between zero and $b-1$, where b is the block size. The low-density parity-check matrix H is recovered from the reduced matrix $H_{\text{sub.rep}}$ by replacing each number $r \in (H_{\text{sub.rep}})_{\text{sub.ij}}$ by the identity matrix of dimension b shifted cyclically r bits to the right and by replacing the empty set by the square zero matrix of dimension b . A quasi-cyclic generator matrix G in standard form may be found for the LDPC code by Gaussian elimination on the blocks. This allows fast encoding of the data bits as explained in [Z. Li, L. Chen, L. Zeng, W. Fong and S. Lin, Efficient encoding of quasi-cyclic low-density parity-check codes, IEEE Trans. Inform. Theory, 54 (2006) 71-81; doi:10.1109/TCOMM.2005.861667].

[0006] A finite incidence structure is given by a pair (P, L) where P is a finite set called the set of points and L is the set of lines, which are subsets of P of a fixed size. Classical examples may be obtained by considering P to be a set of r -dimensional subspaces of a finite dimensional vector space V over a finite field F , and L a set of s -dimensional subspaces of V , where $s \geq r+1$. A point $x \in P$ is incident with the line l (i.e., $x \in l$) if and only if the r -dimensional subspace x is contained in the s -dimensional subspace l . A check matrix for an LDPC code may be made from (P, L) by

labelling the rows by elements x of P and the columns by elements l of L . The corresponding entry is zero if x is not in l and 1 if $x \in l$.

[0007] It is known that finite projective, affine and polar spaces may provide check matrices for LDPC codes which are quasi-cyclic. In [Y. Kou, S. Lin and M. P. C. Fossorier, "Low-density parity-check codes based on finite geometries: a rediscovery and new results," in IEEE Transactions on Information Theory, vol. 47, no. 7, pp. 2711-2736, November 2001, doi: 10.1109/18.959255] the quasi-cyclic structure of finite projective and affine spaces are used to construct efficient LDPC codes of lengths up to approximately 16,000 bits. To construct longer codes however, they use column splitting operations. This results in codes of significantly longer length, up to approximately 500,000 bits, but these codes are not geometric, although they may perform close to Shannon's limit, see Example 4. Meanwhile, in [Zhenyu Liu and D. A. Pados, "LDPC codes from generalized polygons," in IEEE Transactions on Information Theory, vol. 51, no. 11, pp. 3890-3898, November 2005, doi: 10.1109/TIT.2005.856936.] the block sizes b of the quasi-cyclic matrices for finite generalized polygons is determined. Again, however, there is no method given to actually construct these matrices for codes of arbitrarily long length such as, e.g., above approximately 20000 bits. Thus, there is, until now, no known method to calculate the check matrices of long geometric LDPC codes. Herein, it is described how this may be done by using certain equations which allow the construction of a quasi-cyclic check matrix for such codes.

[0008] An object of this disclosure is to provide new methods, systems, computer systems and computer programs aimed at improving currently known digital communication coding methods, systems, computer systems and computer programs.

SUMMARY

[0009] In an aspect, digital communication coding methods (or simply coding methods) are provided, said methods comprising providing families of quasi-cyclic low-density parity-check codes, LDPC codes, each represented by a single equation. Said single equation is derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes and allows quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and exceeding four hundred thousand bits. The construction of such check matrices is not possible by any other known method. The LDPC codes have different lengths and different rates and the families of LDPC codes are provided with a guarantee of minimum distance. The geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates.

[0010] Digital communication coding methods may further comprise puncturing at least one check bit from one or more of the LDPC codes.

[0011] Digital communication encoding methods (or simply encoding methods) are provided. These encoding methods are described in other parts of the disclosure. The encoding is done by adding redundancy bits based on LDPC codes generated by the coding method.

[0012] Digital communication decoding methods (or simply decoding methods) are provided. These decoding methods are described in other parts of the disclosure. The decoding is done by correcting errors introduced by potentially noisy communication channel, said correction based on innovative LDPC codes generated or outputted by the coding method.

[0013] Methods of producing a digital communication coding hardware (or simply coding hardware) may be provided. Said methods include a coding method such as the ones described in other parts of the disclosure and producing the coding hardware based on innovative LDPC codes generated or outputted by the coding method. Digital communication coding hardware generated by methods of producing coding hardware may also be provided.

[0014] In a further aspect, digital communication coding systems (or simply coding systems) are provided. These systems comprise a system module configured to provide families of quasi-cyclic low-density parity-check, LDPC, codes each represented by a single equation. Such a single equation is derived from a geometry whose nature implies good error correcting properties for each

of the LDPC codes and allows quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and in excess of four hundred thousand bits. The construction of such check matrices is not possible by any other known method. The LDPC codes have different lengths and different rates, and the families of LDPC codes are provided with a guarantee of minimum distance. The geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates.

[0015] Digital communication transmission systems (or simply transmission systems) are disclosed. These transmission systems may include an encoding module and a transmission module. The encoding module may be configured to encode data based on innovative LDPC codes generated by any of the methods, systems, computing systems and computing programs aimed at that, which are described in other parts of the disclosure. The transmission module may be configured to transmit the encoded data through, e.g., a potentially noisy communication channel.

[0016] Digital communication receiver systems (or simply receiver systems) are disclosed. These receiver systems may comprise a receiving module and a decoding module. The receiving module may be configured to receive data through, e.g., a potentially noisy communication channel. The decoding module may be configured to decode the received data based on innovative LDPC codes generated by any of the methods, systems, computing systems and computing programs aimed at that, which are described in other parts of the present disclosure.

[0017] In a still further aspect, a computer program is provided comprising program instructions for causing a computer or computing system to perform digital communication coding methods such as the ones described in other parts of the disclosure. The computer program may be embodied on a storage medium and/or carried on a carrier signal.

[0018] In a furthermore aspect, a computing system is provided for coding data in digital communications, the computing system comprising a memory and a processor, embodying instructions stored in the memory and executable by the processor, and the instructions comprising functionality or functionalities to execute digital communication coding methods, such as the ones described in other parts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Non-limiting examples of the disclosure will be described in the following, with reference to the appended drawings, in which:

[0020] FIG. 1 is a block diagram schematically illustrating a digital communication coding system along with systems for producing digital communication coding hardware and software and combination thereof.

[0021] FIG. 2 is a block diagram schematically illustrating digital communication transmission system and digital communication receiver system configured to digitally communicate with each other.

[0022] FIG. 3 is a flow chart schematically illustrating a digital communication coding method along with methods of producing digital communication coding hardware and software and combination thereof, which are suitable for being performed by systems according to, e.g., FIG. 1.

[0023] FIG. 4 is a flow chart schematically illustrating a digital communication encoding method and a digital communication decoding method, which may be used or implemented by digital communication transmitter and receiver systems according to, e.g., FIG. 2.

[0024] FIG. 5 is a flow chart schematically illustrating particular manners of generating LDPC codes according to principles described with reference to FIG. 1.

[0025] FIG. 6 is a schematic graphic representation of the quasi-cyclic check matrix H and the quasi-cyclic generator matrix G for geometry $Q(5,3)$.

DETAILED DESCRIPTION

[0026] FIG. 1 is a block diagram schematically illustrating a digital communication coding system **100** along with systems for producing digital communication coding hardware **101** or software **102** or combination thereof **103**.

[0027] As shown in this figure, digital communication coding system **100** outputs LDPC codes **105** which may then be used by system (also denominated hardware generator system) **101** to produce digital communication coding hardware, that may be configured to generate or produce corresponding digital communication coding hardware (e.g., a chip). This output of the hardware generator system **101** may be or include, e.g., a design or specs of a fully hardware-based implementation of data encoder/decoder based on innovative LDPC codes **105** outputted by digital communication coding system **100**.

[0028] LDPC codes **105** outputted by digital communication coding system **100** may further or alternatively be used by system (also denominated software generator system) **102** to produce digital communication coding software, which may be configured to generate or produce corresponding digital communication coding software (or computer program). This output of the software generator system **102** may be or include a fully software-based implementation of data encoder/decoder based on innovative LDPC codes **105** from digital communication coding system **100**.

[0029] LDPC codes **105** outputted by digital communication coding system **100** may further or alternatively be used by system (also denominated hardware-software generator system) **103** to produce digital communication coding hardware-software, which may be configured to generate or produce corresponding digital communication coding hardware-software. This output of the hardware-software generator system **103** may be or include an implementation of data encoder/decoder combining hardware and software based on innovative LDPC codes **105** from digital communication coding system **100**.

[0030] Digital communication coding systems **100** may comprise a module (also denominated codes generator module) **104** configured to provide families of quasi-cyclic low-density parity-check, LDPC codes **105** each of which is represented by a single equation. Said single equation is derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes **105**. Moreover, the equations provide a quasi-cyclic check matrix for geometric LDPC codes. The lengths of these codes can be exceedingly long. Construction of such check matrices is not possible by any other known method different from digital communication coding systems **100** according to present disclosure. The LDPC codes **105** have different lengths and different rates and the families of LDPC codes **105** have a guarantee of minimum distance. The geometric and quasi-cyclic properties of the LDPC codes **105** allow the implementation of low complexity decoding with very low frame error rates. Different manners of how module **104** in digital communication coding system **100** may operate to generate corresponding LDPC codes **105** are described in detail below.

[0031] A key aspect of module **104** in digital communication coding system **100** is the generation of subsets $(H_{sup.rep})_{sub.ij}$ for geometric LDPC codes **105**. These LDPC codes **105** have good error-correcting properties and are efficient to encode and, in turn, may be decoded quickly using low complexity decoding algorithms. The subsets $(H_{sup.rep})_{sub.ij}$ are predetermined by fixing a finite geometry to which an equation is associated. This equation is a low-degree polynomial $f(x_{sub.i}, a_{sub.j})$ or $f(x_{sub.i}, a_{sub.j}, c_{sub.j})$, where $x_{sub.i}$ (respectively $a_{sub.j}$ or $(a_{sub.j}, c_{sub.j})$) runs through a determined subset of the elements of a finite field $F=GF(q_{sup.h})$ representing a block of points (respectively lines) of the geometry. The block size b is a divisor of $q_{sup.h}-1$. Construction of the geometry in this way allows the removal of small substructures of the geometry that greatly increase the size of the blocks b , thereby enhancing the ease of implementation of the LDPC codes **105**.

[0032] The LDPC codes **105** may therefore be described by their parity check matrix which may be

constructed from the equation, where q may be chosen so that the LDPC code 105 has the desired rate and length. Once q and the geometry have been chosen, the block size b is given in following Table 1, as are the index sets of rows and columns of $H_{sup.rep}$. The rows are indexed by $x_{sub.0}$ and the columns by $a_{sub.0}$ (in the projective, elliptic and symplectic case) and $(a_{sub.0}, c)$ (in the hermitian case), which satisfy the equation given in the final column of following Table 1. Except for the projective case, for each $x_{sub.0}$ and $a_{sub.0}$ satisfying the defining conditions, $x_{sub.i}, a_{sub.j} \in F$ are found such that $x_{sub.i}.sup.b = x_{sub.0}$ and $a_{sub.j}.sup.b = a_{sub.0}$. In the projective case, $x_{sub.i}, a_{sub.j} \in F$ are found such that $x_{sub.i}.sup.b = x_{sub.0}$ and $a_{sub.j}.sup.q.sup.2.sup.+1 = a_{sub.0}$.

TABLE-US-00001 TABLE 1 block dimension finite defining equation condition Geometry size b
 $H_{sup.rep}$ field F $f(x, a)$ or $f(x, a, c)$ on $x_{sub.0}$ and $a_{sub.0}$ $PG(3, q)$ $(q+1)(q^{sup.2}+1)1 \times q$
 $GF(q)$ $yax_{sup.q+1} + x + a$
 $1 - x_{sub.0} = 0$ projective γ fixed such that $\gamma_{sup.q+1} = -1$
 $(\gamma a_{sub.0})_{sup.q} + \gamma a_{sub.0} - 1 = 0$ $x_{sub.0} = x_{sup.b}, a_{sub.0} = a$
 $W(3, q)$ $q^{sup.2} + 1 (q+1) \times q$ $GF(q)$
 $ax_{sup.q+1} + x - \gamma$ $1 -$ $= 0$ symplectic $1 -$ $= 0$ γ fixed such that $= -\gamma$ $x_{sub.0} = x_{sup.b}, a_{sub.0} = a_{sup.b}$ $Q(5, q)$ $q^{sup.3} + 1 (q+1) \times q^{sup.2}$ $GF(q)$
 $ax_{sup.q+1} - x + a$ $1 +$
 $x_{sub.0} +$ $= 0$ elliptic $1 + a_{sub.0} +$
 $+ = 0$
 $x_{sub.0} = x_{sup.b}, a_{sub.0} = a_{sup.b}$ $H(4, q^{sup.2})$ hermitian
 $[00001]$ $(q+1)(q^{sup.2}+1) \times (q+1)(q^{sup.3}+1)$ $GF(q)$
 $1 +$
 $= 0$ condition (2) below $x_{sub.0} = x_{sup.b}, a_{sub.0} = a_{sup.b}$
The block size, dimension of $H_{sup.rep}$, finite field and the defining equation. The rows of $H_{sup.rep}$ are indexed by $x_{sub.0}$ and the columns of $H_{sup.rep}$ are indexed by $a_{sub.0}$ (or $(a_{sub.0}, c)$). $$ indicates data missing or illegible when filed
[0033] The conditions on $a_{sub.0}$ and c for $H(4, q^{sup.2})$ are given by $(c,$
 $a_{sub.0}) \in L_{sub.1} \cup L_{sub.2}$, where:

$$[00002] L_1 \{(c, a_0) \text{ .Math. } F_1(c) \neq 0, F_2(c) = 0, a_0 = \frac{c(c^{q^2+q^4} + c^{q^2} - c^{q^4+q^3-q} - c^{q^4+q^2})}{(c^{q^4+q} + c^4 + c^{q^4+q^3+q} + c^{q^4+q^3} + c^{q^3+q^4})}\} \quad (2) \text{ and}$$

$$F_1(c) = (1+c)^{q^2+1} + c^q, \text{ and } F_2(c) = (1+c)^{q^4+1} + c^{q^4-q^2+1}.$$

[0034] The subset $(H_{sup.rep})_{sub.ij}$ is then determined by the defining equation in following way.

[0035] Let α be a primitive b -th root of unity in the finite field F with $q^{sup.h}$ elements.

[0036] If the defining equation is $f(x_{sub.i}, a_{sub.j})$ (respectively $f(x_{sub.i}, a_{sub.j}, c_{sub.j})$) then the condition is $r \in (H_{sup.rep})_{sub.ij}$ if and only if

$$[00003] f(\alpha^r x_i, \alpha^j) = 0, (\text{resp. } f(\alpha^r x_i, \alpha^j \cdot c_j) = 0) \quad (1)$$

[0037] where r runs from 0 to $b-1$.

[0038] The reduced parity check matrix $H_{sup.rep}$ for the dual of these codes is constructed by the rule

$$[00004] (H_{sup.rep})_{ij} = \{-r \pmod{b} \text{ .Math. } r \in (H^{rep})_{ji}\}.$$

[0039] The length of the code, a lower bound on minimum distance, transmission rate and complexity of constructing $H_{sup.rep}$ are given in following Table 2.

TABLE-US-00002 TABLE 2 block length column row complex - complex - min. approx. Code size b n weight weight $$ $$ dist. rate $PG(3, q)$ $(q+1)(q^{sup.3}+1)qb$ $q+1$ $q(q+1)$ $O(q^{sup.4})$ $O(q$

$\geq q + 2 \cdot 1 - q_{\text{sup.}-1} W(3, q) q_{\text{sup.}2} + 1 q b q + 1 q$
 $O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) \geq 2q \cdot 1 - q$
 $(q \text{ even}) 0.5(q \text{ odd}) W(3, q) q_{\text{sup.}2} + 1 (q + 1)b q q + 1$
 $O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) \geq 2q \cdot 1 - q$
 $(q \text{ even}) \text{dual } 0.5(q \text{ odd}) Q(5, q) q_{\text{sup.}3} + 1 q_{\text{sup.}2} b q + 1$
 $q_{\text{sup.}2} O(q_{\text{sup.}6}) O(q_{\text{sup.}6}) O(q_{\text{sup.}6}) \geq 2q \cdot 1 - q_{\text{sup.}-1} Q(5, q) q_{\text{sup.}3} + 1$
 $(q + 1)b q_{\text{sup.}2} q O(q_{\text{sup.}6}) O(q_{\text{sup.}6}) O(q_{\text{sup.}6}) \geq 2q_{\text{sup.}3} q_{\text{sup.}-1} \text{dual}$
 $H(4, q_{\text{sup.}2}) [00005]_{\text{++?}} (q + 1)(q_{\text{sup.}3} + 1)b q_{\text{sup.}2} + 1 q_{\text{sup.}3} + 1 O(q_{\text{sup.}4})$
 $O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) \geq 2q$
 $1 - q_{\text{sup.}-1} H(4, q_{\text{sup.}2}) \text{dual } [00006]_{\text{++?}} (q + 1)(q_{\text{sup.}2} + 1)b q_{\text{sup.}3} + 1 q_{\text{sup.}2} + 1 O(q_{\text{sup.}4})$
 $O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) O(q_{\text{sup.}4}) \geq q_{\text{sup.}-1} \text{Block}$
 size, length, and complexity of constructing $H_{\text{sup.} \text{rep}}$ and $P_{\text{sup.} \text{rep}}$.

[0040] As mentioned above a quasi-cyclic generator matrix for the LDPC code **105** may be obtained efficiently by performing Gaussian elimination on the blocks of $H_{\text{sup.} \text{rep}}$, to obtain a quasi-cyclic parity check matrix in standard form (which may not be low-density) and then transposing the part of the matrix corresponding to the check bits in the standard way. The quasi-cyclic generator matrix may be described by a reduced generator matrix $P_{\text{sup.} \text{rep}}$ which consists of the first row of each block matrix making up the blocks of the check bit part of the generator matrix. The matrix $P_{\text{sup.} \text{rep}}$ may be constructed as described in [Z. Li, L. Chen, L. Zeng, W. Fong and S. Lin, "Efficient encoding of quasi-cyclic low-density parity-check codes", IEEE Trans. Inform. Theory, 54 (2006) 71-81; doi:10.1109/TCOMM.2005.861667], where examples of circuits are given which implement encodings using such quasi-cyclic generator matrices. The complexity of constructing $P_{\text{sup.} \text{rep}}$ is given in previous Table 2.

[0041] A real-time field programmable gate array (FPGA) decoder was used to implement the LDPC code **105** from $Q(5, 13)$ of length 371462. The performance of this code, as well as computer simulations for shorter length code derived from $Q(5, q)$ (see following Table 3) consistently perform within 1-1.5 dB's of Shannon's limit. Importantly, the LDPC codes **105** have no visible error floor. Another important advantage of using these geometric LDPC codes **105** is that low complexity decoding algorithms may be implemented which perform almost as well as slower sum-product decoding algorithms which require many iterations to converge. Such low complexity decoding algorithms are described in [Y. Kou, S. Lin and M. P. C. Fossorier, "Low-density parity-check codes based on finite geometries: a rediscovery and new results," in IEEE Transactions on Information Theory, vol. 47, no. 7, pp. 2711-2736, November 2001, doi: 10.1109/18.959255]. These algorithms work well partly because of a relatively high column weight. The column weights and row weights of H are detailed in previous Table 2.

TABLE-US-00003 TABLE 3 Elliptic quadric quadrangle $Q(5, q)$. q length

q	length	block size b	$H_{\text{sup.} \text{rep}}$ rank	H dim	C rate	C dim
3	252	28	4	9	91	161
4	555	5	4	4	1040	65
5	779	5	16	261	779	0.74904
6	14405	6	25	651	2499	0.79333
7	14405	6	25	651	2499	0.79333
8	14405	6	25	651	2499	0.79333
9	14405	6	25	651	2499	0.79333
10	14405	6	25	651	2499	0.79333
11	14405	6	25	651	2499	0.79333
12	14405	6	25	651	2499	0.79333
13	14405	6	25	651	2499	0.79333
14	14405	6	25	651	2499	0.79333
15	14405	6	25	651	2499	0.79333
16	14405	6	25	651	2499	0.79333
17	14405	6	25	651	2499	0.79333
18	14405	6	25	651	2499	0.79333
19	14405	6	25	651	2499	0.79333
20	14405	6	25	651	2499	0.79333
21	14405	6	25	651	2499	0.79333
22	14405	6	25	651	2499	0.79333
23	14405	6	25	651	2499	0.79333
24	14405	6	25	651	2499	0.79333
25	14405	6	25	651	2499	0.79333
26	14405	6	25	651	2499	0.79333
27	14405	6	25	651	2499	0.79333
28	14405	6	25	651	2499	0.79333
29	14405	6	25	651	2499	0.79333
30	14405	6	25	651	2499	0.79333
31	14405	6	25	651	2499	0.79333
32	14405	6	25	651	2499	0.79333
33	14405	6	25	651	2499	0.79333
34	14405	6	25	651	2499	0.79333
35	14405	6	25	651	2499	0.79333
36	14405	6	25	651	2499	0.79333
37	14405	6	25	651	2499	0.79333
38	14405	6	25	651	2499	0.79333
39	14405	6	25	651	2499	0.79333
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77	14405	6	25	651	2499	0.79333
78	14405	6	25	651	2499	0.79333
79	14405	6	25	651	2499	0.79333
80	14405	6	25	651	2499	0.79333
81	14405	6	25	651	2499	0.79333
82	14405	6	25	651	2499	0.79333
83	14405	6	25	651	2499	0.79333
84	14405	6	25	651	2499	0.79333
85	14405	6	25	651	2499	0.79333
86	14405	6	25	651	2499	0.79333
87	14405	6	25	651	2499	0.79333
88	14405	6	25	651	2499	0.79333
89	14405	6	25	651	2499	0.79333
90	14405	6	25	651	2499	0.79333
91	14405	6	25	651	2499	0.79333
92	14405	6	25	651	2499	0.79333
93	14405	6	25	651	2499	0.79333
94	14405	6	25	651	2499	0.79333
95	14405	6	25	651	2499	0.79333
96	14405	6	25	651	2499	0.79333
97	14405	6	25	651	2499	0.79333
98	14405	6	25	651	2499	0.79333
99	14405	6	25	651	2499	0.79333
100	14405	6	25	651	2499	0.79333
101	14405	6	25	651	2499	0.79333
102	14405	6	25	651	2499	0.79333
103	14405	6	25	651	2499	0.79333
104	14405	6	25	651	2499	0.79333
105	14405	6	25	651	2499	0.79333
106	14405	6	25	651	2499	0.79333
107	14405	6	25	651	2499	0.79333
108	14405	6	25	651	2499	0.79333
109	14405	6	25	651	2499	0.79333
110	14405	6	25	651	2499	0.79333
111	14405	6	25	651	2499	0.79333
112	14405	6	25	651	2499	0.79333
113	14405	6	25	651	2499	0.79333
114	14405	6	25	651	2499	0.79333
115	14405	6	25	651	2499	0.79333
116	14405	6	25	651	2499	0.79333
117	14405	6	25	651	2499	0.79333
118	14405	6	25	651	2499	0.79333
119	14405	6	25	651	2499	0.79333
120	14405	6	25	651	2499	0.79333
121	14405	6	25	651	2499	0.79333
122	14405	6	25	651	2499	0.79333
123	14405	6	25	651	2499	0.79333
124	14405	6	25	651	2499	0.79333
125	14405	6	25	651	2499	0.79333
126	14405	6	25	651	2499	0.79333
127	14405	6	25	651	2499	0.79333
128	14405	6	25	651	2499	0.79333
129	14405	6	25	651	2499	0.79333
130	14405	6	25	651	2499	0.79333
131	14405	6	25	651	2499	0.79333
132	14405	6	25	651	2499	0.79333
133	14405	6	25	651	2499	0.79333
134	14405	6	25	651	2499	0.79333
135	14405	6	25	651	2499	0.79333
136	14405	6	25	651	2499	0.79333
137	14405	6	25	651	2499	0.79333
138	14405	6	25	651	2499	0.79333
139	14405	6	25	651	2499	0.79333
140	14405	6	25	651	2499	0.79333
141	14405	6	25	651	2499	0.79333
142	14405	6	25	651	2499	0.79333
143	14405	6	25	651	2499	0.79333
144	14405	6	25	651	2499	0.79333
145	14405	6	25	651	2499	0.79333
146	14405	6	25	651	2499	0.79333
147	14405	6	25	651	2499	0.79333
148	14405	6	25	651	2499	0.79333
149	14405	6	25	651	2499	0.79333
150	14405	6	25	651	2499	0.79333
151	14405	6	25	651	2499	0.79333
152	14405	6	25	651	2499	0.79333
153	14405	6	25	651	2499	0.79333
154	14405	6	25	651	2499	0.79333
155	14405	6	25	651	2499	0.79333
156	14405	6	25	651	2499	0.79333
157	14405	6	25	651	2499	0.79333
158	14405	6	25	651	2499	0.79333
159	14405	6	25	651	2499	0.79333
160	14405	6	25	651	2499	0.79333
161	14405	6	25	651	2499	0.79333
162	14405	6	25	651	2499	0.79333
163	14405	6	25	651	2499	0.79333
164	14405	6	25	651	2499	0.79333
165	14405	6	25	651	2499	0.79333

cycles of length less than 6. This, along with bounds on the minimum distance given in previous Table 2, imply good error-correcting properties.

[0043] The parameters of the LDPC codes **105** for various values of q are given in previous Table 3 and following Tables 4, 5, 6, 7 and 8. It is not necessarily the case that a quasi-cyclic generator matrix in standard form exists for all codes C , so the tables include a code C' which is the largest subcode of C for which a quasi-cyclic generator matrix in standard form has been calculated.

TABLE-US-00004 TABLE 4 Dual elliptic quadric quadrangle $Q(5, q)$. q length

 block size b H.sup.rep rank H dim C rate C dim C

 rate C  G.sup.rep(†) 3
112 28 9×4 91 21 0.1875 21 0.1875 1×4 5 756 126 25 $\times 6$ 651 105 0.13889 105 0.13889 1×6
7 2752 344 49×8 2451 301 0.10938 301 0.10938 1×8 9 7300 730 81×10 6643 657 0.09 657
0.09 1×10 13 30772 2198 169 $\times 14$ 28731 2041 0.0663 2041 0.0663 1×14 (†) Since the
dimension of C is less than the block size a non-standard quasi cyclic generator matrix G .sup.rep
may be obtained.  indicates data missing or illegible when
filed

TABLE-US-00005 TABLE 5 Symplectic quadrangle $W(3, q)$. q length

 block size b H.sup.rep rank H dim C rate C dim C

 rate C  P.sup.rep 5 130
26 6×5 91 39 0.3 26 0.2 1×4 7 350 50 8×7 225 125 0.35714 100 0.28571 2×5 8 520 65 9×8
297 223 0.42885 195 0.375 3×5 9 738 82 10×9 451 287 0.38889 246 0.33333 3×6 11 1342
122 12×11 793 549 0.40909 488 0.36364 4×7 13 2210 170 14×13 1275 935 0.42308 680
0.30770 4×9 16 4412 257 17×16 1889 2223 0.54061 1799 0.4375 7×9 17 4930 290 18×17
2755 2175 0.44118 2030 0.41177 7×10 19 6878 362 20×19 3801 3077 0.44737 2896 0.42105
 8×11 23 12190 530 24×23 6625 5565 0.45652 5300 0.43478 10×13 25 15650 636 26×25
8451 7199 0.46 6886 0.44 11×14 27 19710 730 28×27 10585 9125 0.46296 8030 0.40741 $11 \times$
 16 29 24418 842 30×29 13051 11367 0.46552 10104 0.41379 12×17 31 29822 962 32×31
15873 13949 0.46774 13468 0.45161 14×17 32 32800 1025 33×32 12249 20551 0.62655 15375
0.46875 15×17 37 50690 1370 38×37 26715 23975 0.47297 21920 0.43243 16×21 41 68962
1682 42×41 36163 32799 0.52439 31958 0.46341 19×22 
indicates data missing or illegible when filed

TABLE-US-00006 TABLE 6 Dual symplectic quadrangle $W(3, q)$. q length

 block size b H.sup.rep rank H dim C rate C dim C

 rate C  P.sup.rep 4 85 17
 4×5 49 36 0.42353 17 0.2 1×4 5 156 26 5×6 91 65 0.41667 52 0.33333 2×4 7 400 50 7×8
225 175 0.4375 150 0.375 3×5 8 585 65 8×9 297 288 0.49231 260 0.44444 4×5 9 820 82 $9 \times$
 10 451 369 0.45 328 0.4 4×6 11 1464 122 11×12 793 671 0.45833 610 0.41667 5×7 16 4369
257 16×17 1889 2480 0.56764 2056 0.47059 8×9 17 5220 290 17×18 2755 2465 0.47222 2320
0.44444 8×10 19 7240 362 19×20 3801 3439 0.475 3258 0.45 9×11 23 12720 530 23×24
6625 6095 0.47917 5830 0.45833 11×13 25 16276 626 25×26 8451 7825 0.48077 7512 0.46154
 12×14 27 20440 730 27×28 10585 9855 0.48214 9490 0.46429 13×15 29 25260 842 29×30
13051 12209 0.48333 11788 0.46667 14×16 31 30784 962 31×32 15873 14911 0.48438 14430
0.46875 15×17 32 33825 1025 32×33 12249 21576 0.63787 16400 0.48485 16×17 41 70664
1682 41×42 36163 34481 0.51176 33640 0.47619 20×22 
indicates data missing or illegible when filed

TABLE-US-00007 TABLE 7 Hermitian quadrangle $H(4, q$.sup.2). q .sup.2 length

 block size b H.sup.rep rank H dim C rate C dim C

 rate C  P.sup.rep 4 297 11
 15×27 120 177 0.59596 176 0.59259 16×11 9 6832 61 40×112 1891 4941 0.72321 4941
0.72321 81×31 16 66625 205 85×325 11050 55575 0.83415 55350 0.83077 270×5
 25 393876 521 156×756 68251 325625 0.82672 325625

0.82672 625 × 131 text missing or illegible when filed indicates data missing or illegible when filed

TABLE-US-00008 TABLE 8 Dual Hermitian quadrangle H(4, q.sup.2). q.sup.2 length

block size b H.sup.rep rank H dim C rate C dim C

rate C P.sup.rep 4 165 11

27 × 15 120 45 0.27273 44 0.26667 4 × 11 9 2440 61 112 × 40 1891 549 0.225 549 0.225 9 × 31

16 17425 205 325 × 85 11050 6375 0.36585 6150 0.35294 30 × 5

25 81276 521 756 × 150 68251 13025 0.16026 13025

0.16026 25 × 31 indicates data missing or illegible when filed

[0044] Particular manners of generating LDPC codes based on above principles described regarding FIG. 1 will be described later with reference to FIG. 5.

[0045] FIG. 2 is a block diagram schematically illustrating digital communication transmission system 200 and digital communication receiver system 208 configured to digitally communicate with each other. As shown in this figure, transmission system 200 and receiver system 208 may communicate with each other through corresponding channel 204 which may (usually) add noise to the transmission.

[0046] Transmission systems 200 may include an encoding module (or data encoder) 202 and a transmission module 203. Data encoder 202 may be configured to encode data based on innovative LDPC codes 105 generated by digital communication coding systems 100 according to the present disclosure, such as the ones described with respect to FIG. 1. Data encoder 202 may use (or include or implement) any of the methods, systems, computing systems, software, hardware and software-hardware combinations described herein that are configured to encode data (to be transmitted) based on the LDPC codes 105. Transmission module 203 may include, e.g., a modulator configured to modulate encoded data (outputted by encoding module 202) according to, e.g., specs of the channel 204 through which said encoded data is to be transmitted. Transmission systems 200 may further include, as shown in the figure, a source module 201 which may be configured to generate the data to be transmitted, pre-process preexisting data (to be transmitted) prior to its encoding (at data encoder 202), etc.

[0047] Receiver systems 208 may include a receiving module 205 and a decoding module (or data decoder) 206. Receiving module 205 may include, e.g., a demodulator configured to demodulate encoded data received through channel 204 according to, e.g., specs of the channel 204. Data decoder 206 may be configured to decode the received data (optionally once demodulated by receiving module 205) by correcting errors introduced by the (potentially) noisy communication channel 204, based on the innovative LDPC codes 105 generated by digital communication coding systems 100 according to the present disclosure, such as the ones described with respect to FIG. 1. Data decoder 206 may use (or include or implement) any of the methods, systems, computing systems, software, hardware and software-hardware combinations described herein that are configured to decode data (received through channel 204) based on the LDPC codes 105. Receiver systems 208 may further include, as shown in the figure, a sink module 207 which may correspond to a destination unit where decoded data is to be processed according to intended application.

[0048] As used herein, the term “module” or “unit” may be understood to refer to software, firmware, hardware and/or various combinations thereof. It is noted that the modules are exemplary. The modules may be combined, integrated, separated, and/or duplicated to support various applications. Also, a function described herein as being performed by a particular module may be performed by one or more other modules and/or by one or more other devices instead of or in addition to the function performed by the described particular module.

[0049] The modules may be implemented across multiple devices, associated or linked to corresponding digital communication coding systems proposed herein, and/or to other components that may be local or remote to one another. Additionally, the modules may be moved from one

device and added to another device, and/or may be included in both devices, associated to corresponding digital communication coding systems proposed herein. Any software implementations may be tangibly embodied in one or more storage media, such as, e.g., a memory device, a floppy disk, a compact disk (CD), a digital versatile disk (DVD), or other devices that may store computer code or computer instructions.

[0050] The digital communication coding systems according to present disclosure may be implemented by computing means, electronic means or a combination thereof. The computing means may be a set of instructions (e.g., a computer program or software) and then digital communication coding systems may comprise a memory and a processor, embodying said set of instructions stored in the memory and executable by the processor. These instructions may comprise functionality or functionalities to execute corresponding digital communication coding methods such as, e.g., the ones described with reference to other figures.

[0051] In case the digital communication coding systems are implemented only by electronic means, a controller of the system may be, for example, a CPLD (Complex Programmable Logic Device), an FPGA (Field Programmable Gate Array) or an ASIC (Application-Specific Integrated Circuit).

[0052] In case the digital communication coding systems are a combination of electronic and computing means, the computing means may be a set of instructions (e.g., a computer program or software) and the electronic means may be any electronic circuit capable of implementing corresponding digital communication coding methods proposed herein, such as the ones described with reference to other figures.

[0053] The computer program(s) may be embodied on a storage medium (for example, a CD-ROM, a DVD, a USB drive, a computer memory or a read-only memory) or carried on a carrier signal (for example, on an electrical or optical carrier signal).

[0054] The computer program(s) may be in the form of source code, object code, a code intermediate source and object code such as in partially compiled form, or in any other form suitable for use in implementing digital communication coding methods according to present disclosure. The carrier may be any entity or device capable of carrying the computer program(s).

[0055] For example, the carrier may comprise a storage medium, such as a ROM, for example a CD ROM or a semiconductor ROM, or a magnetic recording medium, for example a hard disk. Further, the carrier may be a transmissible carrier such as an electrical or optical signal, which may be conveyed via electrical or optical cable or by radio or other means.

[0056] When the computer program(s) is/are embodied in a signal that may be conveyed directly by a cable or other device or means, the carrier may be constituted by such cable or other device or means. Alternatively, the carrier may be an integrated circuit in which the computer program(s) is/are embedded, the integrated circuit being adapted for performing, or for use in the performance of, digital communication coding methods proposed herein.

[0057] FIG. 3 is a flow chart schematically illustrating a digital communication coding method (or simply coding method) along with methods of producing digital communication coding hardware (or simply hardware producer) and software (or simply software producer) and combination thereof (or simply hardware-software producer), which are suitable for being performed by systems according to, e.g., FIG. 1. Therefore, number references from FIG. 1 may be reused in following description of FIG. 3.

[0058] FIG. 3 thus refers to both coding methods (performed by, e.g., digital communication coding systems **100**) and producer methods (performed by, e.g., digital communication coding systems **100** in cooperation with hardware generators **101** and/or software generators **102** and/or hardware-software generators **103**). Full methods according to FIG. 3 may also be denominated coding and producing methods herein.

[0059] As shown in the figure, coding and producing methods may be initiated (e.g., at block **300**) upon detection of a starting condition such as, e.g., a user request to start the coding and producing

method.

[0060] Coding and producing methods may further include (e.g., at method block **301**) generating LDPC codes **105** in any of the innovative manners described in other parts of the disclosure. This functionality implemented or implementable by method block **301** may be performed by, e.g., codes generator module **104** (in, e.g., digital communication coding system **100**) described with reference to FIG. **1**. Functional details and considerations explained about said codes generator module **104** (in, e.g., digital communication coding system **100**) may thus be similarly attributed or attributable to method block **301**.

[0061] Coding and producing methods may further include (e.g., at method block **302**) verifying whether hardware (option H) or software (option S) or hardware-software (option HS) or any combination thereof (option H and/or option S and/or option HS) has or have been selected. Such an option or options (H and/or S and/or HS and/or any combination thereof) may have been provided by competent user or operator. In case of option H, transition to method block **303** may be performed. Additionally or alternatively, in case of option S, transition to method block **304** may be performed. Additionally or alternatively, in case of option HS, transition to method block **305** may be performed.

[0062] Coding and producing methods may further include (e.g., at method block **303**) generating coding hardware based on innovative LDPC codes **105** (from, e.g., method block **301**). This functionality implemented or implementable by method block **303** may be performed by, e.g., hardware generator system **101** described with reference to FIG. **1**. Functional details and considerations explained about said hardware generator system **101** may thus be similarly attributed or attributable to method block **303**.

[0063] Coding and producing methods may further include (e.g., at method block **304**) generating coding software based on innovative LDPC codes **105** (from, e.g., method block **301**). This functionality implemented or implementable by method block **304** may be performed by, e.g., software generator system **102** described with reference to FIG. **1**. Functional details and considerations explained about said software generator system **102** may thus be similarly attributed or attributable to method block **304**.

[0064] Coding and producing methods may further include (e.g., at method block **305**) generating coding hardware-software based on innovative LDPC codes **105** (from, e.g., method block **301**). This functionality implemented or implementable by method block **305** may be performed by, e.g., hardware-software generator system **103** described with reference to FIG. **1**. Functional details and considerations explained about said hardware-software generator system **103** may thus be similarly attributed or attributable to method block **305**.

[0065] Coding and producing methods may further include (e.g., at method block **306**) ending execution of the method upon satisfaction of an ending condition. Such an ending condition may be satisfied when, e.g., hardware generation has been completed (at, e.g., block **303**), and/or software generation has been completed (at, e.g., block **304**), and/or hardware-software generation has been completed (at, e.g., block **305**), and/or user request has been detected requesting termination of the method, etc.

[0066] FIG. **4** is a flow chart schematically illustrating a digital communication encoding method (or simply encoding method) and a digital communication decoding method (or simply decoding method), according to examples, which may be used or implemented by digital communication transmitter and receiver systems according to, e.g., FIG. **2**. Therefore, number references from FIG. **2** may be reused in following description of FIG. **4**. In particular, encoding method **402** may be performed by encoding module **202**, decoding method **405** may be performed by decoding module **206**, transmission method **401-403** may be performed by transmission system **200**, and receiving method **404-406** may be performed by receiver system **208**. Full methods exemplary illustrated by FIG. **4** may be denominated transmitting and receiving methods. As shown in the figure, transmitting and receiving methods may be initiated (e.g., at block **400**) upon detection of a starting

condition such as, e.g., a user request to start the transmitting (and receiving) method.

[0067] Transmitting and receiving methods may further include (e.g., at method block **401**) generating or pre-processing data to be transmitted. Block **401** may process the data in any manner that prepares the data to be encodable at block **402** and subsequently transmittable at block **403**. Block **401** is not necessary if the pre-existing data does not need processing prior to its encoding at block **402**. This functionality implemented or implementable by method block **401** may be performed by, e.g., source module **201** (in, e.g., digital communication transmission system **200**) described with reference to FIG. 2. Functional details and considerations explained about said source module **201** (in, e.g., digital communication transmission system **200**) may thus be similarly attributed or attributable to method block **401**.

[0068] Transmitting and receiving methods may further include (e.g., at method block **402**) encoding the data to be transmitted based on innovative LDPC codes **105**. This functionality implemented or implementable by method block **402** may be performed by, e.g., encoding module **202** (in, e.g., digital communication transmission system **200**) described with reference to FIG. 2. Functional details and considerations explained about said encoding module **202** (in, e.g., digital communication transmission system **200**) may thus be similarly attributed or attributable to method block **402**.

[0069] Transmitting and receiving methods may further include (e.g., at method block **403**) transmitting the encoded data through potentially noisy channel **204**. This functionality implemented or implementable by method block **403** may be performed by, e.g., transmission module **203** (in, e.g., digital communication transmission system **200**) described with reference to FIG. 2. Functional details and considerations explained about said transmission module **203** (in, e.g., digital communication transmission system **200**) may thus be similarly attributed or attributable to method block **403**.

[0070] Transmitting and receiving methods may further include (e.g., at method block **404**) receiving the encoded data (from transmission system **200**) through a potentially noisy channel **204**. This functionality implemented or implementable by method block **404** may be performed by, e.g., receiving module **205** (in, e.g., digital communication receiver system **208**) described with reference to FIG. 2. Functional details and considerations explained about said receiving module **205** (in, e.g., digital communication receiver system **208**) may thus be similarly attributed or attributable to method block **404**.

[0071] Transmitting and receiving methods may further include (e.g., at method block **405**) decoding the received data based on innovative LDPC codes **105**. This functionality implemented or implementable by method block **405** may be performed by, e.g., decoding module **206** (in, e.g., digital communication receiver system **208**) described with reference to FIG. 2. Functional details and considerations explained about said decoding module **206** (in, e.g., digital communication receiver system **208**) may thus be similarly attributed or attributable to method block **405**.

[0072] Transmitting and receiving methods may further include (e.g., at method block **406**) receiving decoded data at destination unit (in, e.g., receiver system **208**). This functionality implemented or implementable by method block **406** may be performed by, e.g., sink module **207** (in, e.g., digital communication receiver system **208**) described with reference to FIG. 2. Functional details and considerations explained about said sink module **207** (in, e.g., digital communication receiver system **208**) may thus be similarly attributed or attributable to method block **406**.

[0073] Transmitting and receiving methods may further include (e.g., at method block **407**) ending execution of the method upon satisfaction of an ending condition. Such an ending condition may be satisfied when, e.g., decoded data has been received at destination unit, and/or user request has been detected requesting termination of the method, etc.

[0074] FIG. 5 is a flow chart schematically illustrating particular manners of generating LDPC codes according to principles described with reference to FIG. 1. At method block **500**, a suitable geometry is selected based on desired rate and length from Tables 3-8. This determines the

parameters b and q and the finite field F .

[0075] Method block **501** involves running through elements of F to create the index subsets $X_{\text{sub}.0}$ and $A_{\text{sub}.0}$ of F given by the condition on $x_{\text{sub}.0}$ and $a_{\text{sub}.0}$ in the last column of Table 1.

[0076] Method block **502** involves running through elements of F to create the index subsets X and A of F where for each $x_{\text{sub}.0} \in X_{\text{sub}.0}$ an $x \in F$ is found such that $x_{\text{sup}.b} = x_{\text{sub}.0}$ and for each $a_{\text{sub}.0} \in A_{\text{sub}.0}$ an $a \in F$ is found such that $a_{\text{sup}.b} = a_{\text{sub}.0}$. Thus, subsets X and A of F are obtained. The size of these subsets corresponds to the number of rows (respectively columns) of $H_{\text{sup}.rep.}$ (if the geometry $PG(3,q)$ is used then a is found such that $a_{\text{sup}.q.\text{sup}.2.\text{sup}.+1} = a_{\text{sub}.0}$.)

[0077] Method block **503** is the following definition: Let $\alpha = \beta_{\text{sup}.(q.\text{sup}.h.\text{sup}.-1)/b}$, where β is a primitive element of F . In this way, α is a primitive b -th root of unity in F .

[0078] Method block **504** involves the following calculation:

TABLE-US-00009 For i in $\{1, \dots, |X|\}$ do $x_{\text{sub}.i} = X[i]$ for j in $\{1, \dots, |A|\}$ do

 $= A[j]$ $H_{\text{sub}.ij.\text{sup}.rep} := []$ for r in $\{0, \dots, b-1\}$ do

if ( $x_{\text{sub}.i}, a_{\text{sub}.j}$) $= 0$, then

AddSet($H_{\text{sub}.ij.\text{sup}.rep}, r$) endif enddo enddo enddo

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[0079] Method block **505** may include applying Gaussian elimination on the blocks of $H_{\text{sup}.rep.}$, to obtain a quasi-cyclic parity check matrix in standard form (which may not be low-density) and then transposing the part of the matrix corresponding to the check bits in the standard way. The quasi-cyclic generator matrix can be described by a reduced generator matrix $P_{\text{sup}.rep}$ which consists of the first row of each block matrix making up the blocks of the check bit part of the generator matrix.

[0080] FIG. 6 is a schematic graphic representation of the quasi-cyclic check matrix H and the quasi-cyclic generator matrix G , both for geometry $Q(5,3)$, i.e., geometry $Q(5,q)$ with $q=3$.

[0081] The block size is $b=28$ and ω is a primitive element of $F_{\text{sub}.3.\text{sub}.6}$.

[0082] For $q=3$, the following is computed:

$$X = \{\omega_{\text{sup}.2}, \omega_{\text{sup}.474}, \omega_{\text{sup}.616}, \omega_{\text{sup}.572}\}.$$

$$A = \{\omega_{\text{sup}.2}, \omega_{\text{sup}.10}, \omega_{\text{sup}.4}, \omega_{\text{sup}.671}, \omega_{\text{sup}.474}, \omega_{\text{sup}.616}, \omega_{\text{sup}.454}, \omega_{\text{sup}.401}, \omega_{\text{sup}.475}\}.$$

[0083] The matrix H is fully described by $H_{\text{sup}.rep.}$, a $(q+1) \times q_{\text{sup}.2}$ matrix whose rows are indexed by elements of X and whose columns are indexed by elements of A . For $x \in X$ and $a \in A$, the $H_{\text{sub}.a,x}$ entry of $H_{\text{sup}.rep.}$ is a subset $H_{\text{sub}.a,x.\text{sup}.rep}$ of $\{0, \dots, b-1\}$ whose elements indicate a 1-coordinate of the first row of the circulant matrix occupying the corresponding position in the quasi-cyclic matrix H .

[0084] Let $\alpha = \omega_{\text{sup}.q.\text{sup}.3.\text{sup}.-1}$, so that α is a primitive b -th root of unity in $F =$

 $\text{custom-character.sub}.q.\text{sub}.6$.

[0085] The quasi-cyclic representation of the elliptic quadrangle is given by $r \in H_{\text{sub}.x,a.\text{sup}.rep}$ if and only if

$$[00007] a(r x)^{q+1} - r x + a^{q^2} = 0,$$

[0086] where r runs from 0 to $b-1$.

[0087] Returning to the case $q=3$, the quasi-cyclic representation of the elliptic quadrangle $Q(5,3)$ is given by a 4×9 matrix $H_{\text{sup}.rep.}$. To compute the (1,1) entry in $H_{\text{sup}.rep.}$ (i.e. the top-left entry in the matrix), those $r \in \{0, \dots, 27\}$ are determined for which

$$[00008] a(r x)^4 - r x + a^9 = 0,$$

[0088] where $a = \omega_{\text{sup}.2}$, $x = \omega_{\text{sup}.2}$ and $\alpha = \omega_{\text{sup}.26}$. It turns out that no such r exists, so

$H_{\text{sub}.x,a.\text{sup}.rep}$ is the empty set in this case. To determine the next entry in the matrix (i.e. the (1,2) entry), those $r \in \{0, \dots, 27\}$ are computed for which

$$[00009] a(r^4 x^4 - r^9 x + a^9 = 0,$$

[0089] where $a=\omega.\text{sup}.10$, $x=\omega.\text{sup}.2$ and $\alpha=\omega.\text{sup}.26$. It turns out that both $r=9$ and $r=15$ work, so the subset is $\{9,15\}$. Continuing in this way, it is determined that the matrix $H.\text{sup}.rep$ is

$$[00010] \begin{pmatrix} & 9, 15 & 25 & & 8, 12 & 1 & 16 & & 1, 10 \\ 18 & 10 & 0, 18 & 3, 4 & & 7, 19 & 14 & & \\ 1, 9 & 25 & 24 & & 23 & & 14, 16 & 14, 17 & \\ 11 & & & 22, 27 & 15 & 6 & & 0, 15 & 5, 16 \end{pmatrix}.$$

[0090] To construct the quasi-cyclic check matrix H from $H.\text{sup}.rep$, we substitute each empty set with a $b \times b$ matrix of zeros and each subset $\{r\}$ by an identity matrix cyclically shifted to the right by r bits. If there is more than one entry in the subset then we superimpose these cyclic shifts. Thus, the quasi-cyclic check matrix H for $Q(5,3)$ (i.e., with $q=3$) is the one shown in FIG. 6 indicated with number reference **600**.

[0091] Gaussian elimination is applied directly on $H.\text{sup}.rep$, to obtain a quasi-cyclic generator matrix in standard form for the code. As mentioned before, it is not always the case that there exists a generator matrix in standard quasi-cyclic form, so it is provided a generator matrix G for a large subcode C' for which there is a generator matrix in standard quasi-cyclic form. Let k be the dimension of the code C' . The matrix G can be described by a $(k/b) \times ((n-k)/b)$ matrix $P.\text{sup}.repp$, where the (i,j) entry of $P.\text{sup}.rep$ is $P.\text{sub}.ij$, a vector of $\{0,1\}.\text{sup}.b$. It is provided a matrix $P.\text{sup}.rep$, whose entries are the first row of a $b \times b$ circulant matrix. Recall that, replacing each first row with the full circulant $b \times b$ matrix, it is obtained a matrix P , where $G=(P|id)$ is a generator matrix for the implementable code C' .

[0092] In the example for $q=3$ the matrix $P.\text{sup}.rep$ is the following matrix:

TABLE-US-00010 (((0 000000110110101110000001000)(10010010010110101001011100010)(1110101001000010010011110011)(1000000000000000000011011))((000001000110001100000000000000)(1010100100000111111100000101)(1011111001101111111010000111)(000000000000000000000000000000110))((00000010110011101100000000)(011000011001010000000000001000)(010000100101000000001001001)(10000000000000000000000000101101))((00000001110111100000000001000)(1001100101110110011010111010)(1100100000011100110001001110)(00000000000000000000000000000011100))((0000001111100111010000010000)(101100111111001000110100001)(1111001100101100101100101100010111000101100010011111111)(100000000000000000000000011110000)))).

[0093] Thus, the quasi-cyclic generator matrix G for $Q(5,3)$ (i.e., with $q=3$) is the one shown in FIG. 6 indicated with number reference **601**.

[0094] The block size b is 28, so the check matrix H is a 112×252 matrix and the generator matrix G of the code C' is a 140×252 matrix. This $[252,140]$ code C' is the code whose parameters appear in the first row in Table 3. It has been constructed from the quasi-cyclic representation of the elliptic quadric quadrangle $Q(5,3)$. Observe that is fully described by the matrices $H.\text{sup}.rep$ and $P.\text{sup}.rep$.

[0095] It is understood that numerous modifications and variations could be made thereto by those skilled in the art without departing from the spirit and scope of the disclosure. For example, that labelling the rows of a parity check matrix H with r -dimensional subspaces and the columns by s -dimensional subspaces of a certain geometry, where a 1 entry is given by containment, one may deduce equations which if chosen carefully enough a quasi-cyclic matrix may be obtained. More obvious modifications may be made like removing columns blocks or row blocks. It is therefore to be understood that within the scope of the claims, the disclosure may be practiced otherwise than as specifically described herein.

Claims

1. A digital communication coding method, comprising: generating families of quasi-cyclic low-density parity-check, LDPC, codes each represented by a single equation, said single equation being derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes, and the equation allowing quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and exceeding four hundred thousand bits, the construction of such check matrices being not possible by any other known method; wherein the LDPC codes have different lengths and different rates and the families of LDPC codes are generated with a guarantee of minimum distance; and wherein the geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates.
2. The digital communication coding method according to claim 1, further comprising puncturing at least one check bit from one or more of the LDPC codes.
3. The digital communication coding method according to claim 1, wherein the LDPC codes have length equal to or above 20000 bits.
4. A digital communication encoding method, comprising: the digital communication coding method according to claim 1 to generate LDPC codes; and encoding data by adding redundancy bits based on the generated LDPC codes.
5. A digital communication decoding method, comprising: the digital communication coding method according to claim 1 to generate LDPC codes; and decoding data by correcting errors introduced by a noisy communication channel, said correction based on the generated LDPC codes.
6. A method of producing a digital communication coding hardware, comprising: the digital communication coding method according to claim 1 to generate LDPC codes; and producing the digital communication coding hardware based on the generated LDPC codes.
7. The digital communication coding hardware produced by the method according to claim 6 of producing the digital communication coding hardware.
8. A digital communication coding system, comprising: a system module configured to generate families of quasi-cyclic low-density parity-check, LDPC, codes each represented by a single equation, said single equation being derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes, and the equation allowing quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and exceeding four hundred thousand bits, the construction of such check matrices being not possible by any other known method; wherein the LDPC codes have different lengths and different rates and the families of LDPC codes are generated with a guarantee of minimum distance; and wherein the geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates.
9. A computer program product comprising program instructions for causing a computer or computing system to perform a digital communication coding method comprising: generating families of quasi-cyclic low-density parity-check, LDPC, codes each represented by a single equation, said single equation being derived from a geometry whose nature implies good error correcting properties for each of the LDPC codes, and the equation allowing quasi-cyclic parity check matrices to be constructed for geometric LDPC codes of exceedingly long length, up to and exceeding four hundred thousand bits, the construction of such check matrices being not possible by any other known method; wherein the LDPC codes have different lengths and different rates and the families of LDPC codes are generated with a guarantee of minimum distance; and wherein the geometric and quasi-cyclic properties of the LDPC codes allow low complexity decoding with very low frame error rates.
10. The computer program product according to claim 9 embodied on a storage medium.
11. The computer program product according to claim 9 carried on a carrier signal.

12. A computing system comprising a memory and a processor, embodying instructions stored in the memory and executable by the processor, and the instructions comprising functionality or functionalities to execute the digital communication coding method according to claim 1.

13. A digital communication transmitter system, comprising: an encoding module configured to encode data based on LDPC codes generated by the digital communication coding method according to claim 1, or by a digital communication coding hardware based on LDPC codes generated by the digital communication coding method, or by a computer program comprising program instructions for causing a computer or computing system to perform the digital communication coding method, or by a computing system comprising a memory and a processor, embodying instructions stored in the memory and executable by the processor, and the instructions comprising functionality or functionalities to execute a digital communication coding method, and a transmission module configured to transmit the encoded data.

14. A digital communication receiver system, comprising: a receiving module configured to receive data; and a decoding module configured to decode the received data based on LDPC codes generated by the digital communication coding method according to claim 1, or by a digital communication coding hardware based on LDPC codes generated by the digital communication coding method, or by a computer program comprising program instructions for causing a computer or computing system to perform the digital communication coding method, or by a computing system comprising a memory and a processor, embodying instructions stored in the memory and executable by the processor, and the instructions comprising functionality or functionalities to execute the digital communication coding method.
